Daily Review

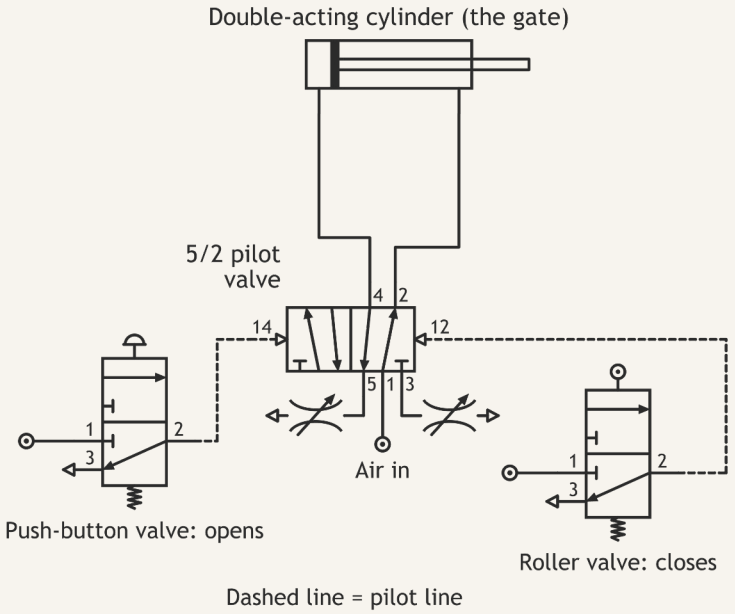


1. Air from either valve gets through: which control?
2. Two valves in a line: which control?
3. Fault-finding: check 1, every time?
4. Double-acting: what brings the rod back?
5. A system diagram: input, process and…

Level Up

The Big Gate

A small push, a big valve, a heavy gate.



S3 Engineering Science

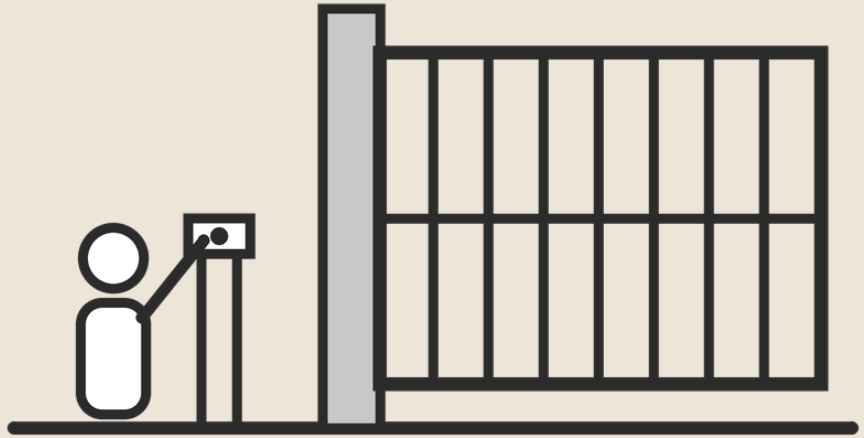
The Job: The Big Gate

i When the button is pressed, the gate opens.

ii When the button is let go, the gate stays open.

iii When the roller is pressed, the gate closes.

Predict: button and roller at once. The gate?



By the end of this lesson…



1. I can build and test the big gate from a bench card
2. I can say what a pilot valve remembers
3. I can read a pilot line and a pilot actuator
4. I can name the motion when a rod goes out and back

Endorsement

Sheet E1 (resit: E2)

3 questions, on your own.
You need 3 out of 3.
Then the kit check with the teacher.

SAFETY No endorsement, no air in Lessons 9 to 11.

2 min

Not Endorsed Yet? Your Route

Tasks 1, 3

build and test it in the sim

Pilot symbol

everyone learns it

Tasks 5-6

the same questions

Resit

sheet E2



THE BUILD

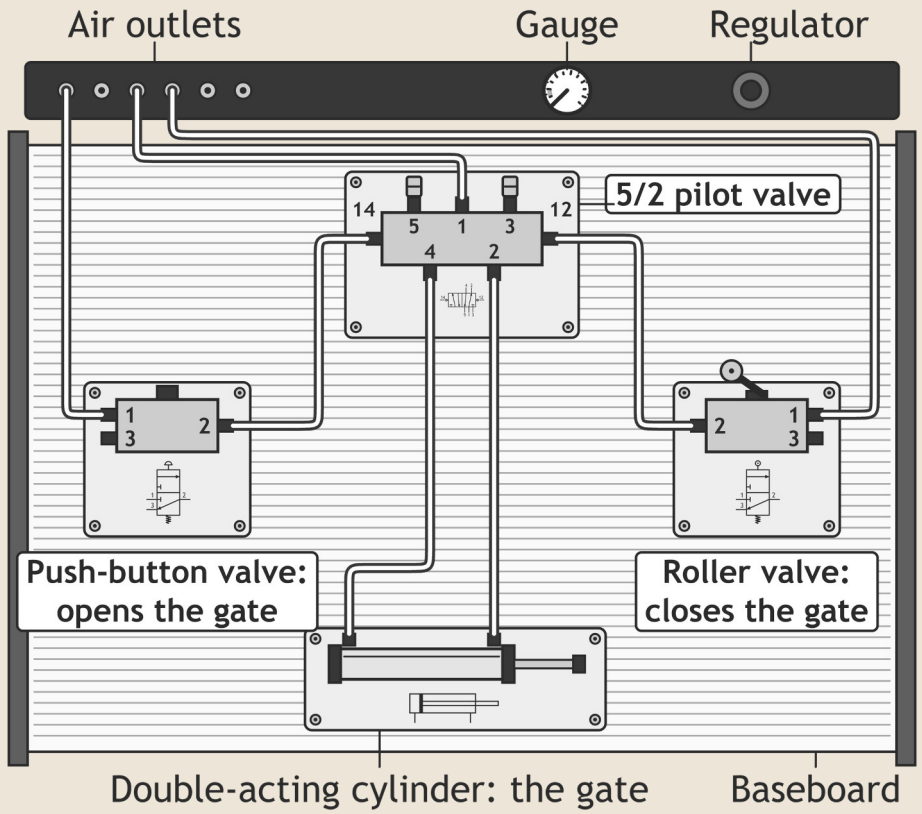
Build the Big Gate

Endorsed? Simulate, Air Permit, then the kit.

What to Collect

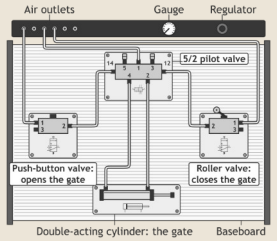
Builder and Safety: one set per team.

1 Bench card 7, tubing
2 Manifold (air supply)
3 5/2 pilot valve
4 Button valve, roller valve
5 Double-acting cylinder
6 2 exhaust restrictors
7 Goggles, one each



Simulate and Build

Tasks 1-2 (B6A-L7-p23)

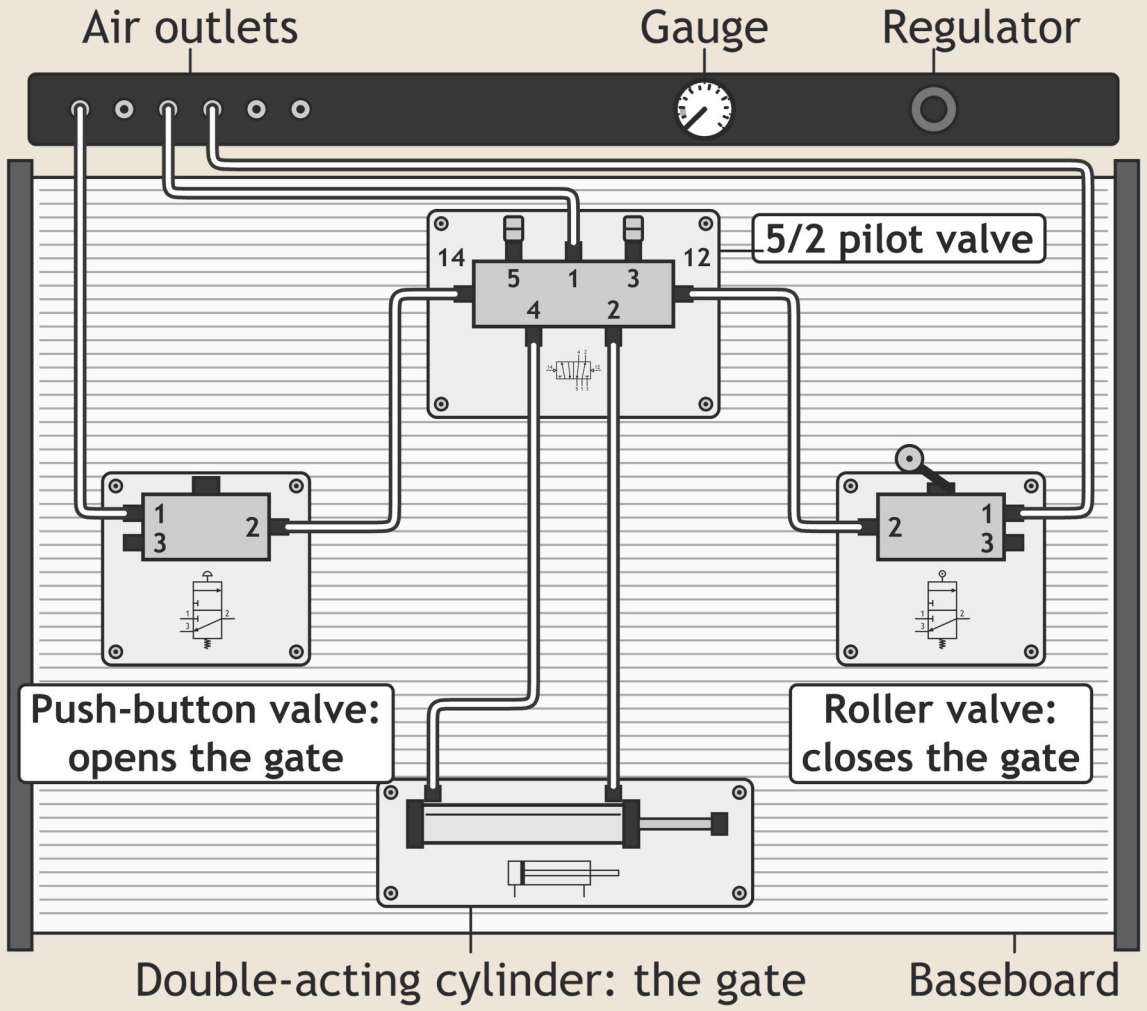


Tester and Recorder: start on the computer.
Builder and Safety: parts and bench card 7.
Safety, then teacher: check the Air Permit.

SAFETY Bolt the roller down first, with the air off.

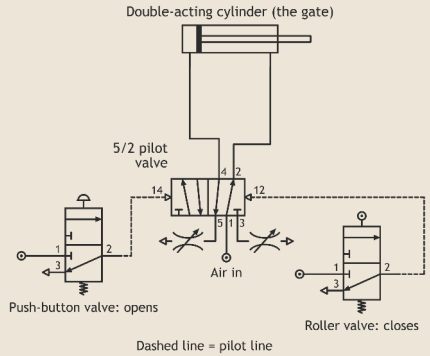
18 min

Build It: Bench Card 7



Find Out: Does It Remember?

Tasks 3-4 (B6A-L7-p23)



Task 3: try every row. Write what the rod does.

Task 4: compare the sim and the kit.

SAFETY Hands clear: the rod can move by itself.

6 min

Test It: What Did You Find? (B6A-L7-p23)



Job	What you do	What the rod does
i, ii	Press the button, then let go.	
iii	Press the roller, then let go.	
-	Press both at once.	
How?	What moved the big valve?	

A Small Push, a Big Valve

Button valve

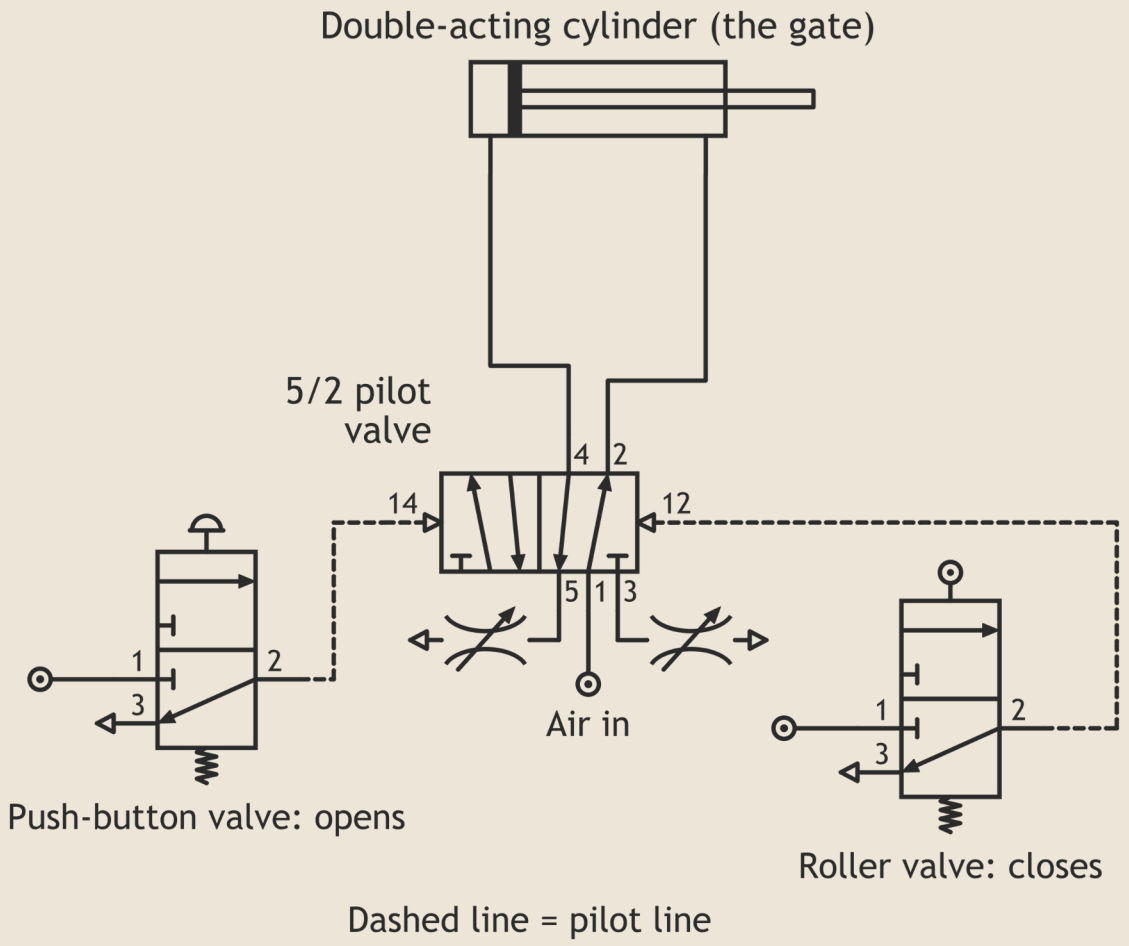
sends a little pilot air

Pilot air

pushes the big valve

Big valve

sends the main air



The Pilot Symbol

Core

Pilot line

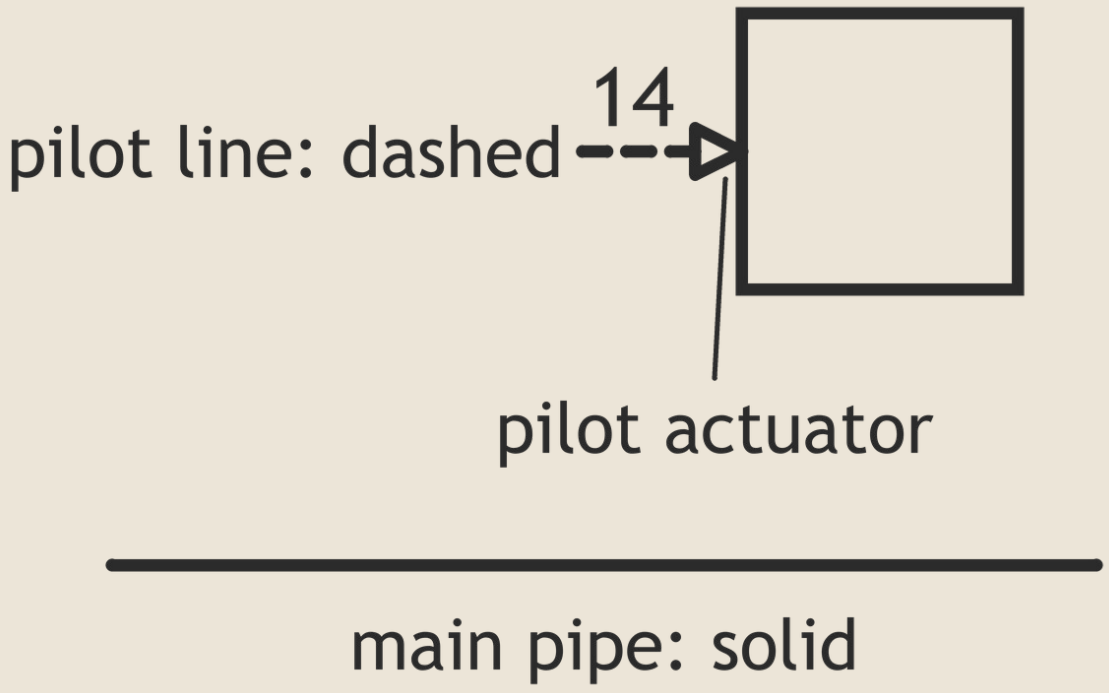
dashed: signal air

Pilot actuator

a triangle at the valve end

Main pipe

solid



The 5/2 Pilot Valve

14

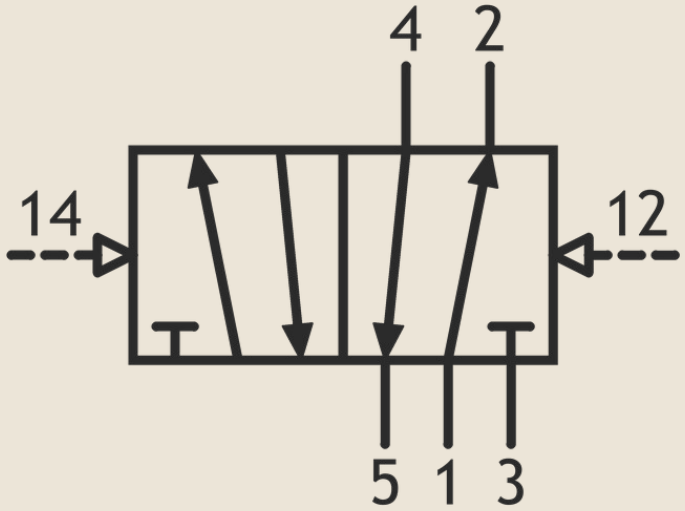
air here: rod goes out

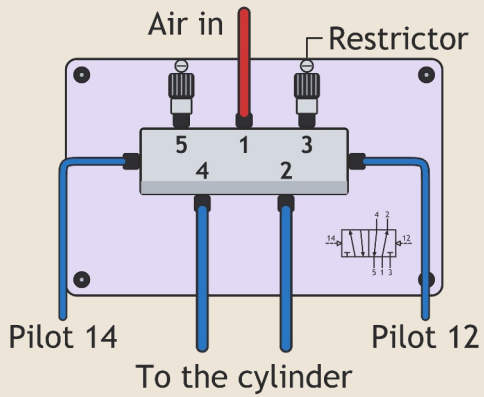
12

air here: rod comes in

No pilot air

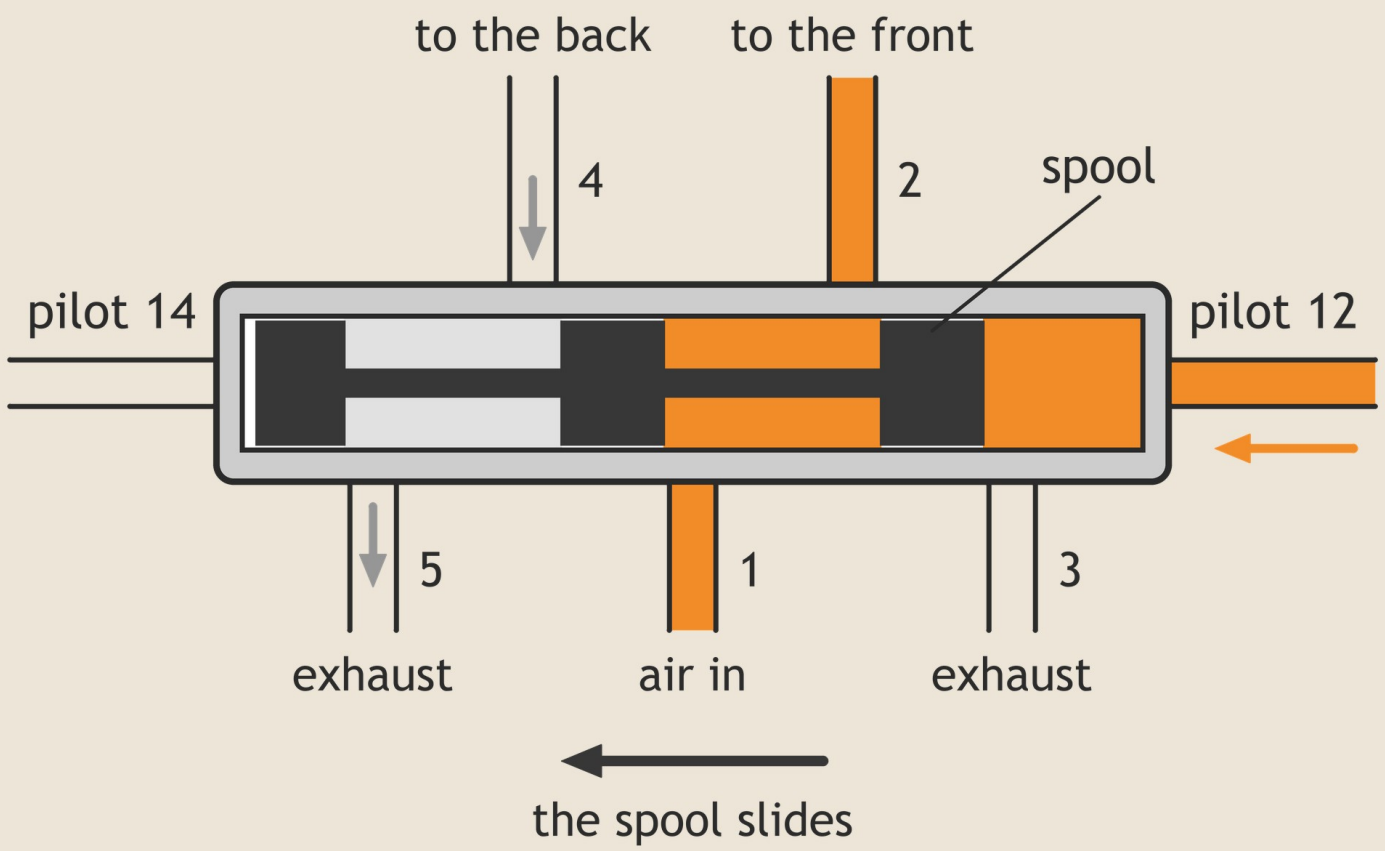
it stays put





The real valve: one per group.

Inside: It Remembers



1 Air to 14 slides the spool. Air 1 to 4: rod out.

2 Let go: no pilot air. The spool stays: rod stays out.

3 Air to 12 slides it back. Air 1 to 2: rod in.

Out and Back: Reciprocating

State the type of motion a pneumatic piston produces when it outstrokes and then instrokes.

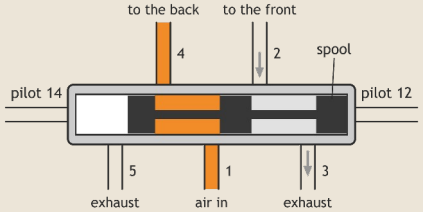
Hint

button: out; roller: back

Motion name	Graphic
Linear	
Reciprocating	

Questions and Key Words

Tasks 5-6 (B6A-L7-p24)



Task 5: Questions 1 to 3 on your own.

Task 6: Key words.

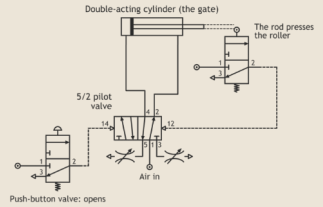
6 min

Answers: Tasks 5-6 (B6A-L7-p24)

Question	Mark 1	Mark 2
Q1 (2 marks)		
Q2 (1 mark)		-
Q3 (1 mark)		-
Key words 1-2		
Key words 3-4		

Optional: A Gate That Closes Itself

Task 8, card 7 side 2



Regulator to 0. Roller to the rod end.
Press the button once. Watch.
Set the restrictors: out fast, back slowly.

SAFETY Move the roller only with the regulator at 0.

8 min

Answers: Task 8 (B6A-L7-p25)

1 Q4 Press the button once. The gate…

Plenary: The Big Gate

1. A pilot line is drawn…
2. Air to pilot port 14: the rod…
3. Let go of the button. The rod…
4. Why can a small button work a big valve?
5. Out, then back: the motion is…

Look back — by the end of this lesson…



1. I can build and test the big gate from a bench card
2. I can say what a pilot valve remembers
3. I can read a pilot line and a pilot actuator
4. I can name the motion when a rod goes out and back

LOOKING AHEAD

Lesson 10: The Pirate Hides by Itself

A pirate pops up, waits, then hides by itself. How?

Credits

The job and the kit: Sources: job card and kit drawings by R Stewart.

Big gate circuit: The big gate circuits adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 5 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Real part drawings: Sources: kit drawings by R Stewart, from the Matrix CP2080 kit.

Out and Back: 2024 Q15(d), 2021 Q1 graphics: © Qualifications Scotland (SQA). N5 Engineering Science 2024, Question 15(d) and its marking instructions; the motion graphics from the 2021 marking instructions, Question 1.

R Stewart - October 2026